

Lesson 4-1: Understand Percent

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Write thirty-seven hundredths as a fraction.

Answer: 37/100

Hundredths means 100 equal parts; 37 are selected.

Prerequisite skill: Read and write a fraction with denominator 100

TASK 2 • FOUNDATION 2

A 10-by-10 grid has 4 complete rows shaded. How many squares are shaded?

Answer: 40

Each row contains 10 squares; $4 \times 10 = 40$.

Prerequisite skill: Count shaded squares on a 100-square grid

TASK 3 • FOUNDATION 3

Write 68/100 as a decimal.

Answer: 0.68

Put 68 in the hundredths places.

Prerequisite skill: Write hundredths as a decimal

TASK 4 • FOUNDATION 4

Simplify 45/100.

Answer: 9/20

Divide the numerator and denominator by their common factor 5.

Prerequisite skill: Simplify a fraction over 100

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TASK 5 • QUICK CHECK A

37 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Answer: Fraction: $37/100$. Decimal: 0.37. Percent: 37%.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

62 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Answer: Fraction $62/100$ (or $31/50$). Decimal 0.62. Percent 62%.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

8 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Answer: Fraction $8/100$ (or $2/25$). Decimal 0.08. Percent 8%.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

What does 150% look like on hundredths grids, and what decimal is it?

Answer: One fully shaded grid plus 50 squares of a second grid — $150/100 = 1.5$.

Check the result against the situation and explain your reasoning.

Lesson 4-2: Relate Fractions, Decimals, and Percentages

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TASK 1 • FOUNDATION 1

Compute 0.36×100 .

Answer: 36

Multiplying by 100 makes each digit worth 100 times as much.

Prerequisite skill: Multiply and divide by 100 by shifting the decimal point

TASK 2 • FOUNDATION 2

Write $3/8$ as a decimal.

Answer: 0.375

Divide 3 by 8; check $0.375 \times 8 = 3$.

Prerequisite skill: Write a fraction as a decimal by dividing

TASK 3 • FOUNDATION 3

Write $3/4$ as a percent.

Answer: 75%

Multiply numerator and denominator by 25 to get $75/100$.

Prerequisite skill: Recall benchmark equivalents for $1/2$, $1/4$, $1/5$, $3/4$, $1/10$

TASK 4 • FOUNDATION 4

Which is greater: 0.6 or 0.56?

Answer: 0.6

Write 0.6 as 0.60; 60 hundredths is greater than 56 hundredths.

Prerequisite skill: Compare decimals to the hundredths place

Lesson 4-2: Relate Fractions, Decimals, and Percentages

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TASK 5 • QUICK CHECK A

0.45 as a percent? 25% as a fraction in simplest form? Which is greater, 0.6 or 55%?

Answer: $0.45 = 45\%$. 25% in simplest form: $25/100 = 1/4$. Comparison: $0.6 = 60\%$, so 0.6 is greater than 55%.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

0.08 as a percent? 40% as a fraction in simplest form? Which is greater, 0.7 or 65%?

Answer: $0.08 = 8\%$. $40\% = 40/100 = 2/5$. $0.7 = 70\%$, so 0.7 is greater than 65%.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

0.9 as a percent? 75% as a fraction in simplest form? Which is greater, 0.3 or 35%?

Answer: $0.9 = 90\%$. $75\% = 75/100 = 3/4$. $35\% = 0.35$, so 35% is greater than 0.3.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Order least to greatest: 0.45, $2/5$, 48%, $1/2$.

Answer: As percents: 45%, 40%, 48%, 50%. Order: $2/5$, 0.45, 48%, $1/2$.

Check the result against the situation and explain your reasoning.

Lesson 4-3: Estimate the Percent of a Number

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TASK 1 • FOUNDATION 1

Find 10% of 350.

Answer: 35

Ten percent is one tenth; $350 \div 10 = 35$.

Prerequisite skill: Find 10% of a number by shifting the decimal one place left

TASK 2 • FOUNDATION 2

Find 1% of 800.

Answer: 8

One percent is one hundredth; $800 \div 100 = 8$.

Prerequisite skill: Find 1% of a number by shifting two places left

TASK 3 • FOUNDATION 3

Find half of 86.

Answer: 43

Halve 80 and 6 separately: $40 + 3 = 43$.

Prerequisite skill: Halve and double a number mentally

TASK 4 • FOUNDATION 4

Estimate 25% of 198 by using 200 as a compatible number.

Answer: 50

25% is one fourth; $200 \div 4 = 50$.

Prerequisite skill: Round to a compatible number that is easy to take a percent of

Lesson 4-3: Estimate the Percent of a Number

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TASK 5 • QUICK CHECK A

10% of 320? 1% of 320? Half of 32?

Answer: 10% of 320 = 32 (shift decimal 1 place left). 1% of 320 = 3.2 (shift decimal 2 places left). Half of 32 = 16 (which represents 5% of 320).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

10% of 480? 1% of 480? Half of 48?

Answer: 10% of 480 = 48. 1% of 480 = 4.8. Half of 48 = 24, which is 5% of 480.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

10% of 250? 1% of 250? Half of 25?

Answer: 10% of 250 = 25. 1% of 250 = 2.5. Half of 25 = 12.5, which is 5% of 250.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Estimate a 15% tip on a \$42.80 bill using benchmarks.

Answer: Use \$40 as a compatible bill amount. 10% is \$4 and 5% is \$2, so a benchmark estimate for a 15% tip is \$6. This is a low estimate because \$40 is less than \$42.80. Other reasonable estimates are possible.

Check the result against the situation and explain your reasoning.

Lesson 4-4: Find and Compare with Percentages

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TASK 1 • FOUNDATION 1

Write 18% as a decimal.

Answer: 0.18

Percent means per hundred; $18 \div 100 = 0.18$.

Prerequisite skill: Write a percent as a decimal

TASK 2 • FOUNDATION 2

Compute 60×0.35 .

Answer: 21

$60 \times 35 \div 100 = 2100 \div 100 = 21$.

Prerequisite skill: Multiply a whole number by a decimal

TASK 3 • FOUNDATION 3

Find $\frac{3}{4}$ of 28.

Answer: 21

Divide 28 by 4, then multiply by 3: $7 \times 3 = 21$.

Prerequisite skill: Multiply a whole number by a fraction

TASK 4 • FOUNDATION 4

Which is greater: 20% of 90 or 30% of 70?

Answer: 30% of 70

The products are 18 and 21; 21 is greater.

Prerequisite skill: Compare two products to decide which is larger

Lesson 4-4: Find and Compare with Percentages

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TASK 5 • QUICK CHECK A

0.25×80 ? $3/4$ of 80? Which is more: 20% of 50 or 30% of 40?

Answer: $0.25 \times 80 = 20$. $3/4$ of 80 = 60.

Comparison: 20% of 50 = 10; 30% of 40 = 12. 30% of 40 is more (12 > 10).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

0.4×90 ? $2/5$ of 90? Which is more: 30% of 60 or 25% of 80?

Answer: $0.4 \times 90 = 36$. $2/5$ of 90 = 36. 30% of 60 = 18; 25% of 80 = 20. So 25% of 80 is more.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

0.6×50 ? $3/5$ of 50? Which is more: 40% of 70 or 50% of 60?

Answer: $0.6 \times 50 = 30$. $3/5$ of 50 = 30. 40% of 70 = 28; 50% of 60 = 30. So 50% of 60 is more.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A \$60 jacket is 30% off. What is the sale price?

Answer: Discount: $0.30 \times 60 = \$18$. **Sale price:** $60 - 18 = \$42$ (or 70% of 60 = \$42).

Check the result against the situation and explain your reasoning.

Lesson 4-5: Determine the Whole Given the Part and Percent

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TASK 1 • FOUNDATION 1

Compute $18 \div 0.3$.

Answer: 60

Scale dividend and divisor by 10: $180 \div 3 = 60$.

Prerequisite skill: Divide a whole number by a decimal

TASK 2 • FOUNDATION 2

Twelve students out of a class of 30 walk to school. Identify the part and the whole.

Answer: Part 12; whole 30

The part counts walkers; the whole counts everyone in the class.

Prerequisite skill: Tell which number is the part and which is the whole

TASK 3 • FOUNDATION 3

Compute $9 \div \frac{1}{4}$.

Answer: 36

There are 4 fourths in each whole, so $9 \times 4 = 36$ fourths.

Prerequisite skill: Divide by a benchmark fraction such as $\frac{1}{4}$ or $\frac{1}{2}$

TASK 4 • FOUNDATION 4

A student says the whole is 80 when 25% is 20. Check by finding 25% of 80.

Answer: 20

$0.25 \times 80 = 20$, so the proposed whole checks.

Prerequisite skill: Check an answer by multiplying back

Lesson 4-5: Determine the Whole Given the Part and Percent

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TASK 5 • QUICK CHECK A

$12 \div 0.25$? 12 is 25% of what number? Now check: 25% of your answer.

Answer: $12 \div 0.25 = 48$. The whole is 48 (12 is 25% of 48). **Check:** 25% of 48 = $0.25 \times 48 = 12$ (balances!).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$15 \div 0.5$? 15 is 50% of what number? Now check: 50% of your answer.

Answer: $15 \div 0.5 = 30$. The whole is 30.
Check: $0.5 \times 30 = 15$. ✓

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$9 \div 0.2$? 9 is 20% of what number? Now check: 20% of your answer.

Answer: $9 \div 0.2 = 45$. The whole is 45. **Check:** $0.2 \times 45 = 9$. ✓

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A team won 18 games, which was 75% of the games played. How many games were played?

Answer: $18 \div 0.75 = 24$ games. **Check:** $0.75 \times 24 = 18$. ✓

Check the result against the situation and explain your reasoning.